The Best Open Source Platforms For Developing BLOCKCHAIN APPLICATIONS

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This article introduces some popular open source platforms that help in quick and easy development of blockchain-based applications

Blockchain technology is termed as the fifth evolution of computing. Blockchain is regarded as a novel approach towards distributed databases. It is a disseminated database of open records that documents all transactions or advanced events that have been executed prior to that point in time when it is shared among participating members. Every transaction in the public record is confirmed by agreement by a larger number of members within that framework.

These days, transactions between persons or companies are often centralised and controlled by a third-party organisation. Making a digital payment or currency transfer requires a bank or credit card provider as the middleman to complete the transaction. In addition, a transaction attracts a fee from the bank or credit card company.

The same process applies to other application domains like games, software and so on. The transaction system is centralised, and all data and information are controlled and managed by a third-party organisation, rather than the two principal participants in the transaction. Blockchain technology has been developed to solve this issue.

Blockchain is a data structure that makes it possible to create a digital ledger of data and share it among a network of independent parties. The ultimate goal of blockchain technology is to create a decentralised environment where no third party is in control of the data and the transactions.

Blockchain is a distributed database solution that maintains a growing list of data records that are confirmed by the nodes participating in it. Data is recorded in a public ledger, which includes all the information about every transaction ever completed. This information is shared and available to all nodes. This, in turn, makes the blockchain system more transparent and leaves no room for a third party.

Blockchain is secure by design and includes sophisticated distributed computing systems with high Byzantine Fault Tolerance.

Blockchain technology was invented by Satoshi Nakamoto in 2008 to serve as a public transaction ledger of the cryptocurrency called Bitcoin. Till date, Bitcoin is the most commonly used application using blockchain.

The invention of blockchain for Bitcoin made it the first digital currency to solve the double spending problem without any need for a centralised server or third-party trusted authority. It is a decentralised digital currency payment system that consists of a public transaction ledger called blockchain.

There are many different kinds of blockchains, explained as follows:

Public blockchain. It is like Bitcoin, a large distributed network that runs through a native token. It is open for the public to participate at any level and has open source code that the community maintains.

Permissioned blockchain. Permissioned blockchain like Ripple controls roles that individuals can play within the network. It is a large and distributed system that uses a native token. Its code may or may not be open source.

Private blockchain. Private blockchain tends to be smaller and does not use a token. Its membership is closely controlled. This type of blockchain is favoured by consortiums that have trusted members and trade in confidential information.

All the above blockchains use cryptography to allow each participant on any given network to manage the ledger in a secure way, without the need for any central authority to enforce the rules.

Underlying principles of blockchain

The following are the five basic principles underlying the technology:

Distributed database. Every participant on the blockchain has access to the entire database and its complete history. No single